Chapter 4
DRUGS ACTING ON GASTROINTESTINAL TRACT

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Jan/2024

Diseases in the GIT

- 1) Constipation
- 2) Diarrhea
- 3) Emesis
- 4) Peptic ulcers and gastroesophageal reflux disease (GERD)

DRUGS ACTING ON GASTROINTESTINAL TRACT

❖LAXATIVES/PURGATIVES

- Are Rx for constipation
  - ➢ by Accelerating stool transit through the intestines.

- Constipation is infrequent bowel movement (≤3 stools/week)

- Laxatives are classified by their mechanism of action

1. *BULK FORMING LAXATIVES*

- These agents are polysaccharide polymers that are not digested
- Indigestible, hydrophilic Colloids that absorb water,
- form a bulky gel that cause water retention, distends the colon and promotes peristalsis.
- Natural: ***psyllium, methylcellulose***
- Synthetic fiber: ***polycarbophil***
- **S/E-** Bacterial digestion of the fiber in colon→ bloating and flatus
  - may cause intestinal obstruction (use cautiously in immobile patients)

GIT cont'd

2. ***STOOL SOFTNERS /SURFACTANT/***

- Soften stool material
- Allow water + lipid to penetrate stools
- this has an emulsifying effect on the feces, making them retain more water and hence softer - easier to pass out
- Docusate sodium/calcium, liquid paraffin
- **S/E-** Absorbs the fat soluble vitamins in the gut, and therefore you loose your fat soluble vitamins

3. LUBRICANT LAXATIVES

- *Mineral oil* and *glycerin suppositories*
- are lubricants that act by facilitating the passage of hard stools
- These are oily laxatives that coat the intestines to help move stool through quicker.
- S/E- can cause lipid pneumonia if aspiration occurs.

4. *OSMOTIC LAXATIVES*

- Consist of poorly absorbed solutes resulting in increased stool liquidity
- It produce an osmotic load then trap increased volumes of fluid in the lumen
- Treatment of acute constipation and prevention of chronic constipation
- ***Nonabsorbable sugar***: sorbitol & lactulose
  - *Lactulose* is a semisynthetic disaccharide sugar
  - S/E-metabolized by colonic bacteria → severe flatus & cramp
  - *Lactulose* is also used for the treatment of hepatic encephalopathy, due to its ability to reduce ammonia levels.
- ***Nonabsorbable salt***: magnesium hydroxide/ milk of magnesia
  - S/E-hypermagnesemia in renal insufficient patient if used for prolonged period
- ***Polyethylene glycol(PEG)***: are used for complete colonic cleansing prior to GI endoscopy
- High dose of osmotically active agents produce purgation within 1-3hs
- S/E- bloating and flatulence, electrolyte imbalance

5. ***STIMULANT LAXATIVES/ CATHARTICS/***

- Direct stimulation of the enteric nervous system and stimulate colonic fluid and electrolyte secretion
- Useful in neurologically impaired and in bed bound patients in long term care facility
- **Anthraquinone derivatives**

  -Aloe, senna & cascara: bowel movement in 6-8hs after p.o. and increase colon secretion(water and electrolyte secretion into the bowel)

  - cause brown pigmentation of colon "melanosis coli"
- **Bisacodyl**- is a potent stimulant of the colon (it acts directly on nerve fibers in the mucosa of the colon)
- **Caster oil**: hydrolyzed in small intestine to ricinoleic acid-irritant that stimulates motility of colon.
- S/E- Long term: dependency & destruction of myenteric plexus; atony and electrolyte imbalance, uterine contraction(*castor oil*)

Contraindication to use of laxatives

- Acute abdominal pain
  - Suspected Intestinal obstruction,
  - Suspected appendicitis,
- Use with caution in elderly and renal failure patients due to high risk of
  - ➢Electrolyte imbalance
  - ➢Dehydration

GIT cont'd

❖TREATMENT OF DIARRHEAS

- Diarrhea: too rapid evacuation of stools (more than 3 times/day)
- Therapeutic measures:
  - treatment of fluid depletion, shock, and acidosis & maintenance of nutrition
  - Use of spasmolytic or other antidiarrheal agents
  - Use of anti-infective agents to treat the underlying cause
- i. **Rehydration**
  - Intravenous rehydration in severe fluid loss
  - Oral rehydration solution (ORS) if the fluid loss is mild
- ii. **Antimotility agents**-like *loperamide* inhibit acetylcholine release in the GIT and decrease peristalsis

iii. Agents that modify fluid and electrolyte transport

- *Bismuth subsalicylate*,
  - ➢used for traveler's diarrhea,
  - ➢decreases fluid secretion in the bowel.
    - ➢By protective coating in the intestinal mucosa
  - ➢Besides its antidiarrheal effect, it can also relieve heartburn
  - ➢Adverse effects may include ***black tongue*** and black stools

- **iv. Antimicrobials** are of no use in diarrhea due to noninfective causes
  - Antimicrobials are useful only in mod-severe GIT infection cases
  - Traveler's diarrhea: can be caused by E. coli, Salmonella, Shigella and Campylobacter pylori [cotrimoxazole, ciprofloxacin, doxycycline, erythromycin]
  - For amoebiasis, giardiasis: metronidazole, tinidazole

❖ANTI EMETICS/ANTI-VOMITING DRUGS

❑Emesis/Vomiting

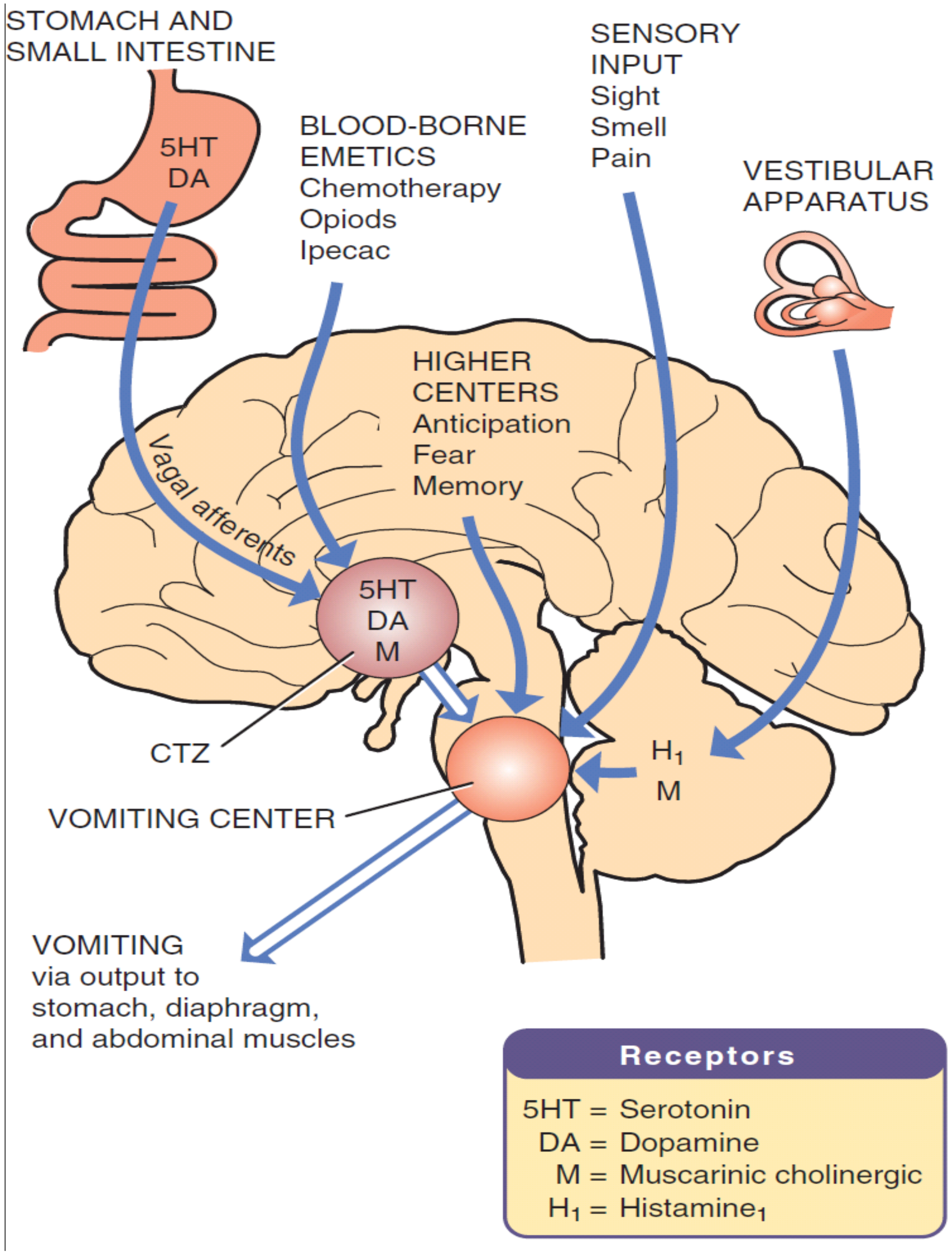
- Is a complex series of integrated events which results in the forceful expulsion of gastric contents through the mouth
- *Is vomiting good or bad?*
- Nausea and vomiting occur in a variety of conditions (for e.g - motion sickness, pregnancy, some drugs etc..)
- Nearly 70% to 80% of patients who take chemotherapy drugs experience nausea and/or vomiting.

Drugs which causes nausea and vomiting

- ➢Apomorphine (and Bromocriptine)
  - ✓Acts on the D2 receptor in the CTZ to cause vomiting
- ➢Cisplatin(anticancer drug)
  - ✓causes release of serotonin in the GIT

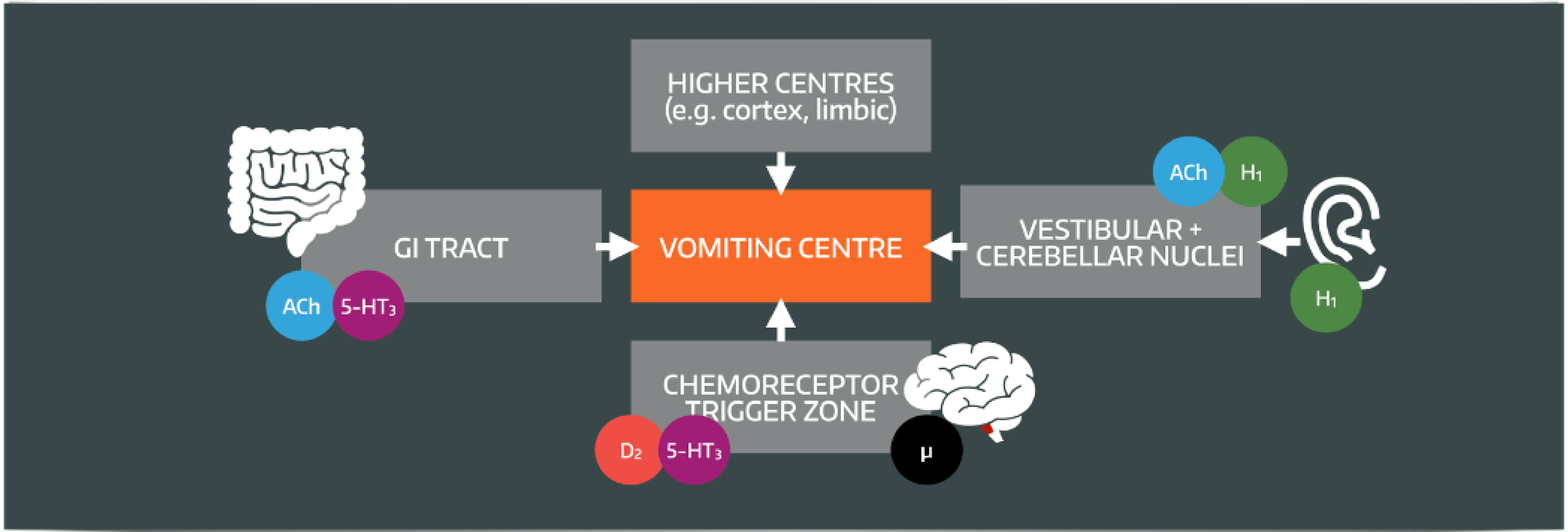
- **Vomiting center-** coordinates the motor part of vomiting
  - Is the main area responsible for vomiting
  - It receives input from many areas:
  - ✓Chemoreceptor trigger zone(CTZ)-picks up circulating chemical in the blood
  - ✓Vestibular apparatus from ears
  - ✓vagal afferents from the GIT
  - ✓pain, smell, fear from brain cortex

- **Receptors involved in the emetic response**
  - On the CTZ: 5HT3(serotonin rec), D2(Dopamine rec)
  - On the vagal afferents from GIT: 5HT3, D2
  - Vestibular Sy: Muscarinic, H1 (Histamine) receptors



Main chemical transmitters & receptors involved in control of vomiting

- **Acetylcholine** – Muscarinic receptors;
- **Histamine**- H_1 Rs;
- **Serotonin** – 5-HT_3 Rs;
- **Dopamine** – D_2 Rs
- **Substance P** - Neurokinin 1 rec


HIGHER CENTRES
(e.g. cortex, limbic)
GI TRACT
ACh
5-HT3
VOMITING CENTRE
ACh
H1
VESTIBULAR +
CEREBELLAR NUCLEI
H1
CHEMORECEPTOR
TRIGGER ZONE
D2
5-HT3
μ

- The chemoreceptor trigger zone (CTZ) is located in the area postrema (around the fourth ventricle). *It is outside the blood–brain barrier*. Thus, it can respond directly to chemical stimuli in the blood or cerebrospinal fluid.
  - rich in dopamine **D 2** receptors and **opioid** receptors, and possibly serotonin 5-HT 3 receptors and neurokinin 1 (NK 1) receptors

- The vomiting center, which is located in the medulla, coordinates the motor mechanisms of vomiting.
  - Receives afferent input from the **vestibular system**, **GI tract**), **CTZ** and **cortical** structures.
  - Rich in muscarinic **M 1** , histamine **H 1** , neurokinin 1 (NK 1), and serotonin 5-HT 3 receptors
- The vestibular system functions mainly in motion sickness.

Antiemetic drugs

- Drugs which decrease or prevent nausea and vomiting

1. H_1 receptor antagonists:

Antihistamines
- H1 blockers- for allergy, flu, motion sickness
- H2 blockers- for gastritis

- ➢ *Block H1 rec and also block M rec*
- ➢ *diphenhydramine, dimenhydrinate, cyclizine, meclizine, promethazine*
- ➢ Are effective in motion sickness
- ➢ *Promethazine and doxylamine* are used in vomiting of pregnancy (morning sickness)
- ➢ S/E- sedation, dry mouth, urinary retention, constipation

2. Anticholinergics

- *Scopolamine* (hyoscine) – used in the prevention and treatment of motion sickness
- Suppresses nerve traffic in neuronal pathway from vestibular apparatus of inner ear to vomiting center
- Mechanisms- block M and also blocks H1 receptors
- S/E- dry mouth, urinary retention, constipation, mydriasis

3. Anti-dopaminergics

I. Antipsychotics

- ***Phenothiazines***: chlorpromazine, perphenazine, prochlorperazine
- ***Butyrophenones***: haloperidol and droperidol

➢ *Act mainly as antagonists of the dopamine D_2 receptors in the CTZ, in addition they also block histamine and muscarinic receptors*

➢ Used for treating nausea and vomiting associated with cancer, radiation therapy, cytotoxic drugs, anesthetics etc

➢ for *severe morning sickness* of pregnancy

S/E- *Parkinsonism like effect(dyskinesia, tremor etc)*

II. Metoclopramide

- Widely used
- is a D_2 receptor antagonist
- Acts centrally on the CTZ and also has a peripheral action on the GIT itself - increasing the motility of the esophagus, stomach and intestine ⇒ prokinetic agents

Therapeutic uses

- Antiemetic in postoperative vomiting, radiation, drug induced vomiting, cancer therapy, GI disorders
- Gastro-esophageal reflux disease
- Gastric hypomotility

S/E- parkinsonism like effect

4. 5-HT_3 receptor antagonists

- ***Ondansetron, granisetron, dolasetron, palonosteron, tropisetron***

- Potent antagonists of 5-HT_3 receptor, primarily in the CTZ; and also on peripheral vagal nerve terminals
- Are the most widely used drugs for chemotherapy-induced emesis; also used to prevent or treat post-operative nausea and vomiting

S/E- headache and constipation

5. Neurokinin 1 (NK 1)-receptor antagonists

- have antiemetic properties that are mediated through central blockade in the area postrema.
- Can be used for chemotherapy induced vomiting
- **Aprepitant** - is a highly selective NK 1 -receptor antagonist that crosses the blood-brain barrier and occupies brain NK 1 receptors. It has no affinity for serotonin, dopamine receptors

DRUGS FOR PEPTIC ULCER Disease (PUD)

- **Ulcer:** Breakdown of the protective mucosal layer
  - ➢Common sites: Duodenum & Stomach
  - ➢Pain is due to:
    - ✓Acid acting on the erosion
    - ✓Increase in the motility of the gut, causing increased intramural tension (antimotility agents decrease the pain)

Gastric mucosa

- The stomach secretes about 2.5 liters of gastric juice daily.
- **Chief or peptic cells-** produce pepsinogen
- **Parietal or oxyntic cells -** produce hydrochloric acid (HCl) and intrinsic factor . Gastric acid is important for promoting proteolytic digestion of foodstuffs, iron absorption and killing pathogens.
- **Mucus-secreting cells -** produce ***bicarbonate*** ions which are secreted and trapped in the ***mucus***, creating a gel-like protective barrier that maintains the gastric mucosal surface.
  - Locally produced 'cytoprotective' **prostaglandins** stimulate the secretion of both mucus and bicarbonate, thus it protect the mucosa from acid
  - Disturbances in these secretory and protective mechanisms are thought to be involved in the pathogenesis of peptic ulcer
  - E.g Alcohol, drugs can disrupt this layer.

Secretion of gastric acid

- Acid (HCl) is secreted from gastric parietal cells/oxyntic cells by a proton pump (H+/K+ ATPase)

Secretion of HCl by parietals enhanced by 3 major endogenous stimuli (activators of proton pump)

- **Acetylcholine** stimulates M -Rec - ↑ Ca^{++} in parietal cells
- **Gastrin** activates gastrin Rec - ↑ Ca^{++} in parietal cells
- **Ach** and **Gastrin** activate release of Histamine from enterochromaffin-like cells (ECL) which stimulates H_2 rec on parietal cells (the main pathway of proton pump activation)

Inhibitors of HCl secretion

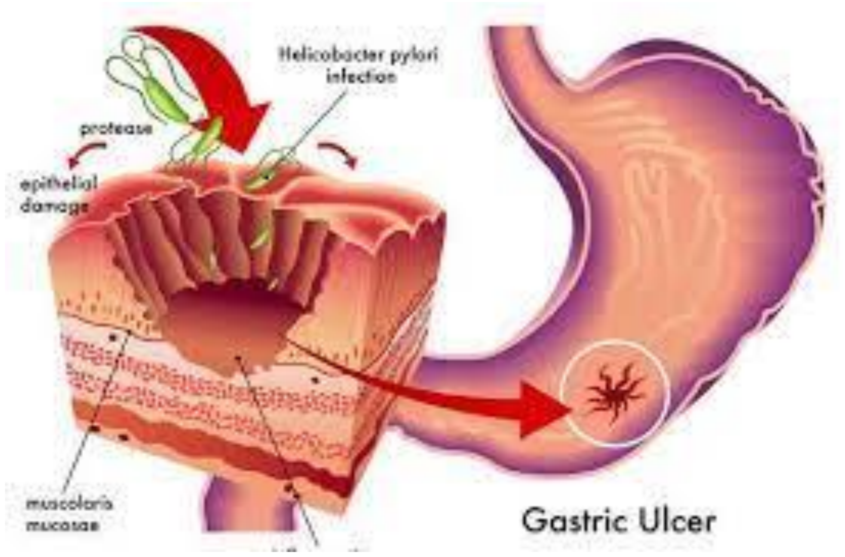
- Somatostatin through somatostatin receptors
  - Released from ***D cells*** at several locations within the stomach. It exerts inhibitory actions on gastrin release from **G cells,** histamine release from ECL cells, as well as directly on parietal cell acid output
- **Prostaglandins E and I** ⇒ inhibit ***gastric acid secretion***, stimulate ***mucus and bicarbonate*** secretion by gastric mucosa; promote vasodilatation & ***maintain blood flow***

Gastric mucosal protection

- Mechanisms that protect the gastric mucosa from autodigestion (damaging effects of gastric acidity and peptic enzymes)
  - Mucus secretion
  - Bicarbonate secretion (surface epithelial cells)
  - Mucosal blood flow
  - Prostaglandin synthesis (favour production of mucus and bicarbonate, inhibit acid secretion by parietal cells, PGE and PGI improve mucosal blood flow)

Common Causes of peptic ulcer disease

- A. *Helicobacter pylori infection* of gastric mucosa



- B. an imbalance between the mucosal-damaging and the mucosal-protecting agents
  - This can occur due to ingestion of some drugs(e.g NSAIDS, alcohol, Khat chewing etc)

Parietal cells in the stomach secrete H+ using *proton pumps*
The secretion of H+ by the pump is regulated by three mechanisms
The effects of each of these mechanisms is cumulative

- Histamine release by ECL cells
- Ach release by PSNS motor neurons
- Gastrin release by G cells

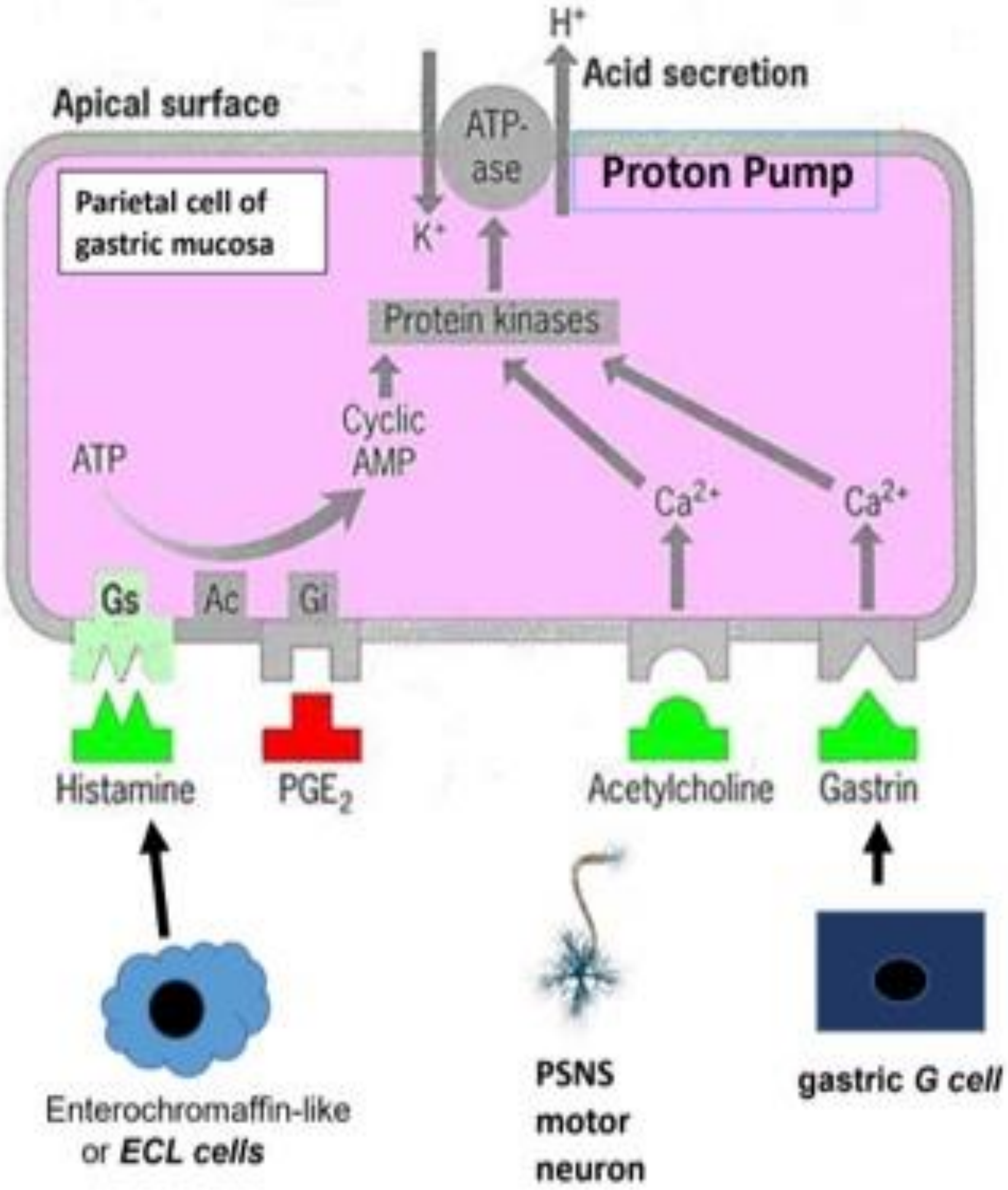
Consider the following drugs used in the treatment of ulcers.
How do they help in the treatment of ulcers?

a. antibiotics

b. antacids

c. histamine blockers

d. proton pump inhibitors



Treatment of PUD

1. Anti-secreting agents

i. H2 antagonists

- Ranitidine, Famotidine, Nizatidine and Cimetidine- prototypical drug
- Mechanism of action:
  - By blocking the H2 receptor → decrease acid secretion
- Little side effect associated with ranitidine
- Allows the ulcer to heal by decreasing gastric acid

- ADR of H2 blockers:
  - CNS- Mental status change (confusion, agitation, hallucination)
  - Endocrine effect: cimetidine inhibits binding of dihydrotestosterone, inhibits metabolism of estradiol, increase prolactin [gynecomastia, impotence in male; galactorrhea in F]
  - Cross the placenta & secreted into breast milk

GIT cont'd

Drug interaction: cimetidine interfere hepatic cytochrome P450 drug metabolism pathways [warfarin, theophylline, phenytoin, lidocaine, quinidine, propranolol, labetalol, metoprolol, TCAs, several benzodiazepines, CCBs, sulfonylureas, metronidazole, and ethanol]

GIT cont'd

ii. M1 selective antagonist

- Spasms of the gut can exacerbate the pain of peptic ulcers.
- Since gut motility is controlled by parasympathetic vagal fibres, anticholinergics would be useful here
- **Pirenzepine** (M1 selective antagonist) can block M1 receptor, thus decrease acid secretion also decrease the spasm of GIT

GIT co

iii. Proton pump inhibitors

- **Omeprazole**, lansoprazole, rabeprazole, pantoprazole & esomeprazole
- They are prodrug, undergoes molecular conversion in parietal cell i.e. reactive thiophilic sulfonamide
- H+ K+ ATPase irreversibly inhibited
- The drugs have short half life but duration of acid inhibition lasts up to 24hs
- Used in :-
    - GERD as first line
    - Peptic ulcer [*H.pylori*- triple therapy=PPI + Clarithromycin + Amoxicillin/metronidazole]
    - Gastrinoma
- Adverse effects: diarrhea, headache, & abdominal pain.

  Reduced B12 release; enteric infection

  Increased serum gastric level

GIT cont'd

2. Antacids

- **Mg** (OH)**2**
  - Causes diarrhea
  - Can also be used as laxatives
  - In excess amounts there will be systemic absorption
- **CaCO3**
  - Causes constipation
- **Al(OH)3**
  - Constipation
  - Al**3+** binds phosphates, leading to hypophosphatemia

- N.B. *combination of antacids* is commonly used. [Mg &Al salts]
- Antacid can impair absorption of tetracyclines

3. *Mucosal protective agents*

i. **Prostaglandin analogues:** MISOPROSTOL

- A synthetic PG-E**2** analogue
- PGE**2** protects the stomach in a number of ways:

- limits the amount of gastric acid being released
- It increases mucous & bicarbonate secretion
- It increases blood flow to the stomach
- Misoprostol[p.o.] can be given in patients taking NSAIDS
- Side effects:
    - ➢Colic and diarrhea
    - ➢Dangerous in pregnancy → PGE**2** contracts the uterus

GIT cont'd

ii. Colloidal bismuth subsalicylate

- Physically *coats the ulcer* site
- Needs an acidic environment to work
- Given as a brown liquid which smells like ammonia therefore, had low patient compliance. [Tablet alternatives]
- It has toxic effects on the **H. pylori**, and may also prevent its adherence to the mucosa or inhibit its bacterial proteolytic enzymes.
- Can be used to eradicate *H. pylori (quadruple therapy)*

iii. Sucralfate

  - Salt of sucrose complexed to sulfated aluminum hydroxide
  - binds to *ulcers or erosions* for up to 6hs

4. Anti-infective agents

- *When the PUD causative agent is **H. pylori**, anti-infectives are used*
- Amoxicillin, Clarithromycin , Metronidazole

- **Eradication of *H.pylori***

- Triple therapy: for 14 days
  - PPI(e.g omeprazole 20mg bid)
  - Clarithromycin 500mg bid
  - Amoxycillin 1gm bid or Metronidazole 500mg bid

- **Quadruple therapy:**
  - triple therapy + bismuth salicylate 525mg bid